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Learn how to make accurate estimates: Calibration training with an interactive online application.

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ABSTRACT

Accurate estimation is a fundamental skill in decision science, essential for effective decision-making across various fields, including agriculture, environmental management, and policy development. To address the challenges of making reliable estimates, we developed an innovative online application designed for calibration training that helps students and professionals enhance their estimation skills using 90% confidence intervals. Our app provides a structured learning environment where users engage with binary and range questions, facilitating the development of their estimation capabilities. Participants are introduced to the concept of confidence intervals and trained to construct them effectively, fostering a deeper understanding of uncertainty in their estimates. We integrate insights into cognitive biases that often impede accurate estimation throughout the training. Participants learn to recognize and mitigate these biases—such as the Dunning-Kruger effect, anchoring bias, and confirmation bias—through practical examples and targeted exercises. By emphasizing the psychological factors influencing decision-making, we aim to cultivate a more nuanced understanding of how biases can skew judgment and lead to suboptimal decisions. Our app features a dual-interface design, catering to participants undergoing calibration and analysts facilitating the training. Participants (including experts, students, and stakeholders) are guided through questions that challenge their estimation skills, while analysts (lecturers, hosts, and administrators) can monitor progress and provide real-time feedback. This dual approach enhances the learning experience and promotes collaborative discussions around estimation techniques.

KEYWORDS: Calibration Training, Confidence Intervals, Decision Science, Cognitive Biases, Estimation Techniques

INTRODUCTION

Accurate estimation is essential in decision science, providing the foundation for effective decision-making across diverse fields such as agriculture, environmental management, and policy development. Precise estimates are critical in agricultural planning, where they guide decisions that impact food security and sustainability (Luedeling et al., 2015), while in policy

development, they support robust risk assessments that shape policies affecting large populations (Hubbard, 2014).

Despite its importance, reliable estimation remains challenging. Cognitive biases—including overconfidence, anchoring, and framing—often skew judgment, leading to estimates that do not accurately reflect actual values (Kahneman et al., 2011). Furthermore, limited training opportunities in uncertainty awareness make it difficult to overcome these biases effectively.

To address these challenges, we developed an interactive online application for calibration training aimed at improving estimation accuracy through structured practice and bias awareness. This app provides a practical, engaging environment where participants can refine their estimation skills, build confidence in producing reliable estimates, and develop a nuanced understanding of uncertainty. By training users to recognize and mitigate biases, our app offers a valuable tool for decision scientists and professionals across fields.

Designed specifically to support decision-making in rural development, this tool fosters collaboration among diverse teams, including farmers, development professionals, and researchers, enhancing their capacity to make informed, effective decisions tailored to agricultural and community contexts.

LITERATURE REVIEW

Accurate estimation facilitates effective decision-making by guiding actions with high stakes, such as resource allocation and policy formation, particularly where the consequences of uncertainty can be substantial (Spetzler et al., 2016). Confidence intervals are a common statistical tool used to represent the range within which a true value is likely to fall, and they are critical to the probabilistic reasoning necessary for complex models, such as those employing Monte Carlo simulations (Morgan & Henrion, 1990; Hubbard, 2014). Despite its importance, reliable estimation is frequently challenged by cognitive biases and gaps in formal training, making calibration training a promising yet complex solution.

Calibration training seeks to improve estimation accuracy by addressing biases and increasing individuals' awareness of their own uncertainty. Hubbard (2014) advocates for calibrated probability assessment as essential for improving decision-making in uncertain conditions, while Lichtenstein et al. (1982) argue that structured feedback in estimation tasks reduces overconfidence and enhances judgment accuracy. However, the effectiveness of calibration training is critical and can be hard to manage. Calibration involves comparing expert-provided probability distributions to actual outcomes. Experts often provide narrower ranges and more accurate estimates than novices, but their tighter ranges do not always encompass the actual outcomes, especially in complex problems (Spence 1996). Structured elicitation protocols aim to mitigate biases and ensure expert judgments are scrutinized similarly to empirical data (Hanea & Nane 2019).

The influence of cognitive biases on estimation accuracy is a critical consideration. For example, individuals with limited expertise often overestimate their own abilities, which can lead to overconfident and unreliable estimates (cf. Kruger & Dunning, 1999). This bias is particularly relevant in decision-making contexts, where individuals may lack formal statistical training and thus remain unaware of their limitations (Miller & Geraci, 2011). Anchoring bias, where initial reference points affect subsequent estimates, further complicates calibration efforts, as initial

exposures can set mental benchmarks that persist despite contradictory evidence (Tversky & Kahneman, 1974; Epley & Gilovich, 2006). Other biases, such as confirmation bias and framing effects, exacerbate these issues by reinforcing existing beliefs and shaping judgment based on information presentation (Nickerson, 1998; Tversky & Kahneman, 1981). These biases reveal that even trained individuals may struggle with estimation accuracy, suggesting that calibration training may not fully overcome these deep-seated cognitive tendencies.

While calibration training has demonstrated utility, its limitations warrant careful consideration. Despite its strengths in structured learning and awareness-building, calibration training can be resource-intensive, requiring iterative practice and significant cognitive engagement, which may not be feasible in all settings (Larrick et al., 1993; Morgan & Henrion, 1990). Additionally, Spetzler et al. (2016) note that while calibration training can improve awareness of uncertainty, it does not necessarily eliminate biases, especially in high-stakes or time-pressured environments where cognitive biases may resurface despite training. Furthermore, evidence suggests that the complexity of the estimation task and contextual factors, such as stress or fatigue, can undermine the effectiveness of calibration (Koriat et al., 1980). Thus, while calibration training is beneficial, it may need to be adapted or supplemented with additional strategies, such as adaptive feedback mechanisms or hybrid approaches that integrate real-time scenario adjustments.

These findings underscore the need for a more nuanced understanding of calibration training's role within decision science. A holistic approach to improving estimation accuracy would benefit from integrating cognitive psychology and adaptive training methods that account for environmental pressures, cognitive load, and the varied nature of biases across individuals. By tailoring training strategies to specific decision contexts and user needs, decision scientists can improve the robustness of estimation models, ultimately leading to more reliable and effective decision-making frameworks across diverse sectors.

Expanding on Expert Engagement and Motivation

One area that remains underexplored in the literature on calibration training is the challenge of maintaining expert engagement over extended periods of estimation tasks. Research suggests that even well-calibrated experts may experience boredom and disengagement when faced with repetitive tasks, particularly when the estimation process involves a lengthy list of variables (Deci & Ryan, 1985). This disengagement can lead to a reduction in accuracy, as experts may be less motivated to apply the same level of rigor to each estimate. Maintaining engagement in these contexts requires consideration of intrinsic and extrinsic motivation, as well as the cognitive demands of the task (Ryan & Deci, 2000).

Motivation plays a crucial role in sustaining the accuracy and quality of expert contributions. According to self-determination theory (Deci & Ryan, 1985), providing opportunities for autonomy, competence, and relatedness can enhance intrinsic motivation, which is essential for sustained engagement. In the context of calibration training, allowing experts to have some level of control over their tasks—such as selecting the order in which they estimate variables or incorporating breaks—may improve engagement. Additionally, providing regular feedback and recognizing the value of their contributions can enhance competence and relatedness, further promoting sustained motivation (Ryan & Deci, 2000; Locke & Latham, 2002).

Moreover, task variety and gamification elements have been suggested as potential strategies to mitigate boredom in repetitive training tasks (Hamari et al., 2014). Introducing challenges,

rewards, or interactive elements could make the process more engaging and help maintain focus, particularly in scenarios requiring prolonged estimation efforts. Such strategies are consistent with findings from educational psychology, which indicate that varying task structure and incorporating fun elements can sustain interest and prevent disengagement (Csikszentmihalyi, 1990).

Adaptability in Calibration Training for Diverse Users

Another critical aspect of calibration training is its adaptability, particularly when working with diverse communities that may include illiterate participants or individuals with limited access to technology. Traditional calibration training methods often assume a level of literacy and technological familiarity that may not be present in all contexts, limiting the effectiveness of such training for certain groups (Rogers, 2003). To address this, calibration training approaches must be adaptable and inclusive, providing alternative pathways for participation.

The need for adaptability is further emphasized by the varied contexts in which decision science operates. In low-income, rural, or technologically constrained environments, training programs must be flexible enough to accommodate different levels of expertise and resource availability (Chambers, 1994). By tailoring calibration training to meet the needs of specific communities, decision scientists can enhance both the inclusivity and the effectiveness of these interventions, ultimately improving decision-making outcomes across diverse sectors.

METHODS

Calibration App Design and Functionality

We developed an interactive online application to train participants in accurately estimating 90% confidence intervals, a foundational skill in decision science. Confidence intervals are essential inputs for decision models, particularly Monte Carlo simulations, which facilitate robust risk assessment and uncertainty quantification (Lanzanova et al 2019; Whitney et al. 2017). The app's training structure draws from Hubbard's (2014) principles of calibrated probability assessment to enhance decision-making precision and is further supported by our own insights on using Monte Carlo simulations for stakeholder-oriented benefit analysis and integrating expert estimates in agricultural decision models.

Our app is built as a Shiny application in R and hosted via R Studio Connect, making it accessible for use in formal lectures, expert workshops, and other decision analysis contexts. It comprises two main components: binary and range questions. This dual-question format allows users to engage with estimation tasks that improve their understanding and execution of confidence intervals, cultivating accuracy in a structured, interactive environment that reduces common cognitive biases.

Access links:

- Binary questions: https://agtools.app/calibration_app_binary/
- Range questions: https://agtools.app/calibration_app_range/
- Code repository: [Zenodo: 10.5281/zenodo.11408263](https://zenodo.org/record/11408263)

The app's user interface contains two primary tabs:

- Questions Tab: Participants input answers on the left, with results displayed on the right (Figure 1). Clicking the "Truth" column in the results table links users to the source for each correct answer.
- Metrics Tab: Displays the user's progress, showing how closely a perfectly calibrated user's estimates align with the desired confidence interval, represented visually by a blue bar meeting a green circle (Figure 2).

Question:3

How many national parks are there in Germany? (in 2021)

90% Confidence Interval

Lower Bound: Upper Bound:

[→ Next](#)

Range Results

Question	90% Confidence Interval		Truth	Relative Error
	Lower Bound	Upper Bound		
1	600	1000	612	-0.94
2	1740	1890	1835	0.27

Figure 1: Screenshot of the "Questions" tab in the calibration app with range questions.

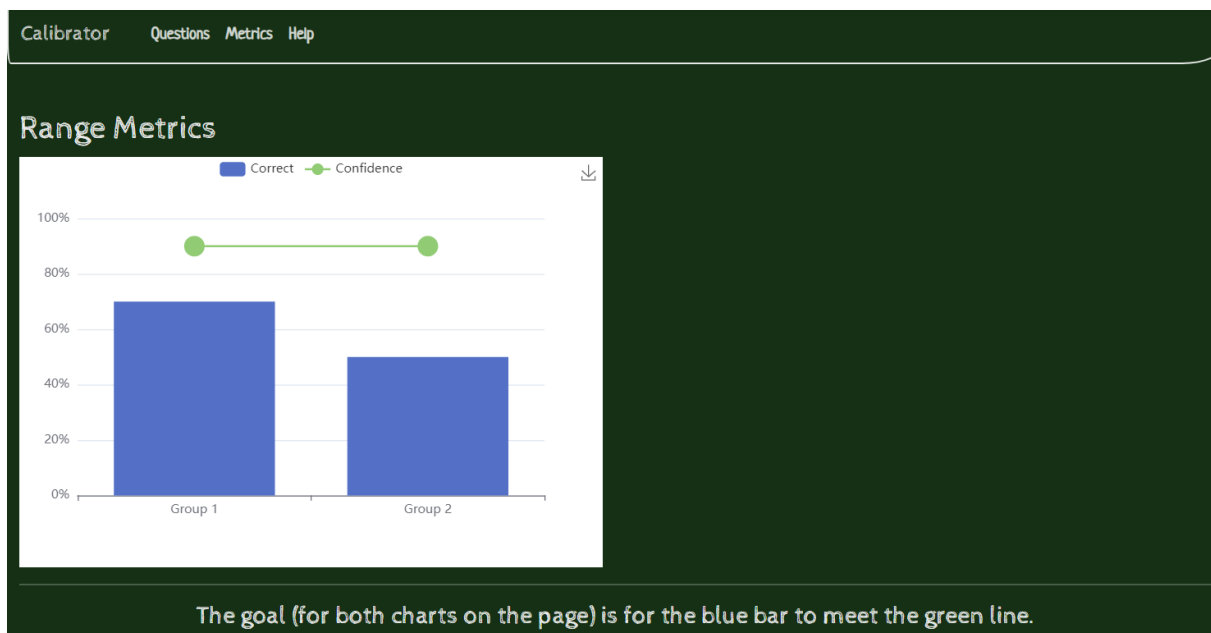


Figure 2: Screenshot of the "Metrics" tab in the calibration app with range questions. This participant answered two blocks of questions.

Real-Time Feedback and Flexibility

A key advantage of the app is its ability to automatically evaluate responses and display progress in real-time, offering immediate feedback to help participants assess strengths and areas for improvement. This enables targeted practice and supports different training formats—whether in-person, hybrid, or online—enhancing accessibility and extending its reach to a variety of learning environments. The app also allows users to repeat question rounds after initial training, reinforcing concepts and promoting ongoing skill development.

Group Calibration and the Admin Interface

The app is complemented by an admin interface that aggregates group responses, enabling real-time group performance analysis. At the end of each question block, the analyst can display group results compared to individual progress, fostering collaborative learning and self-assessment (Figures 2, 3).

Admin app access: https://agtools.app/calibration_admin/
Admin code repository: [Zenodo: 10.5281/zenodo.11408199](https://zenodo.org/record/11408199)

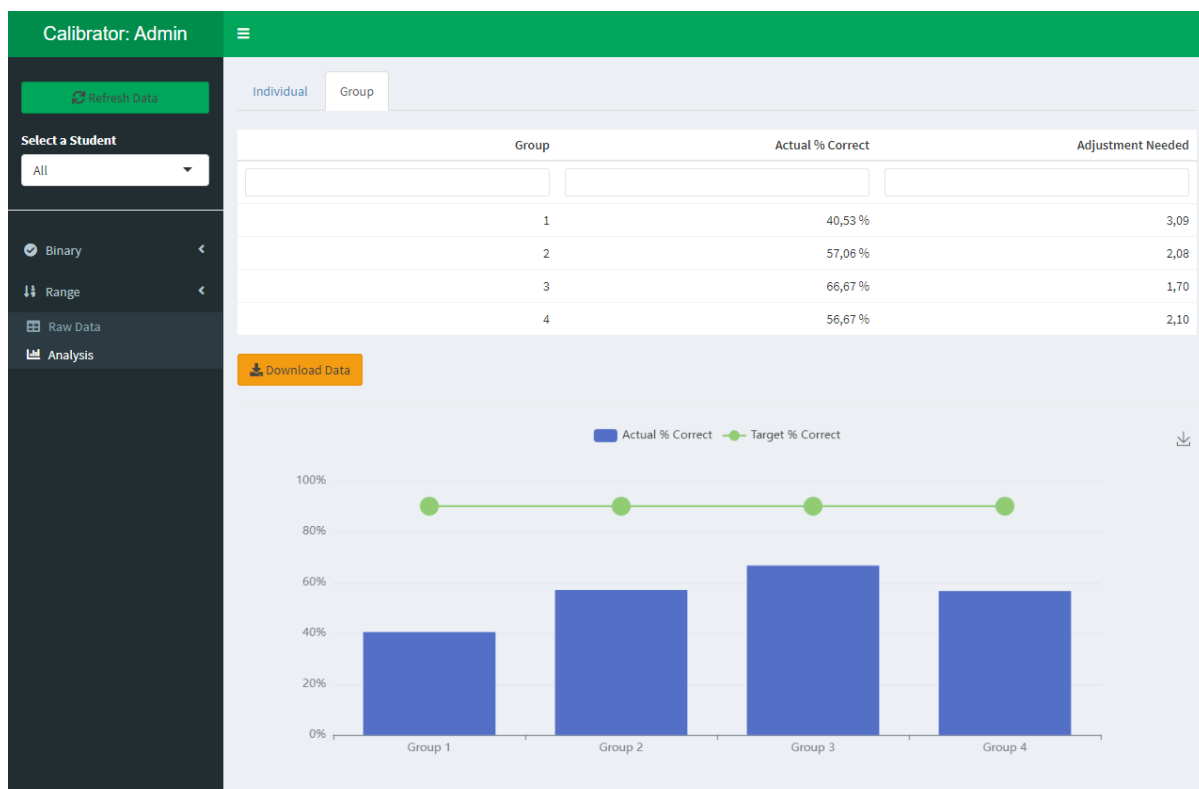


Figure 3: Screenshot of the admin app. Group results of a training with four question blocks of range questions. Note: Not all participants finished blocks three and four.

Training Methodology and Cognitive Bias Awareness

The training introduces participants to 90% confidence intervals and explores the psychological biases affecting judgment, such as the Dunning-Kruger effect, anchoring, and framing effects (Kahneman et al., 2011). Participants are encouraged to mitigate these biases through practical exercises and targeted estimation techniques (Table 2).

Question Types and Training Flow

Training alternates between analyst presentations and participant engagement in the app. Using three to five question blocks, participants begin with binary questions to build initial confidence, followed by range questions to deepen understanding and precision (Figure 4).

- Binary Questions: Participants assess the accuracy of true/false statements with adjustable confidence sliders, reinforcing foundational judgment skills (Figure 6).
- Range Questions: Participants estimate intervals for numerical questions, practicing techniques that enhance their calibration skills over time (Figure 7).

Prerequisites

To participate in the training, every participant must have a stable internet connection and a computer or mobile device with an internet browser. They must also be comfortable with computers and numbers. We have, at times, worked with interpreters to act as an interface for experts who do not meet these requirements.

Key Training Content

Introducing the Calibration App:

The session begins with an introduction to calibration training as a practical tool for improving estimation accuracy and self-awareness of uncertainty. Drawing on the principles of Applied Information Economics (Hubbard, 2014), we emphasize the importance of quantifying uncertainty for effective decision-making. We demonstrate model parameterization using Luedeling et al. (2015) as an example, underscoring how accurate estimates impact complex decision-making outcomes, especially in agriculture. Additionally, we clarify the difference between precision (consistency of estimates) and accuracy (proximity to the true value) (Figure 5), equipping participants with a foundation for calibrated probability assessment.

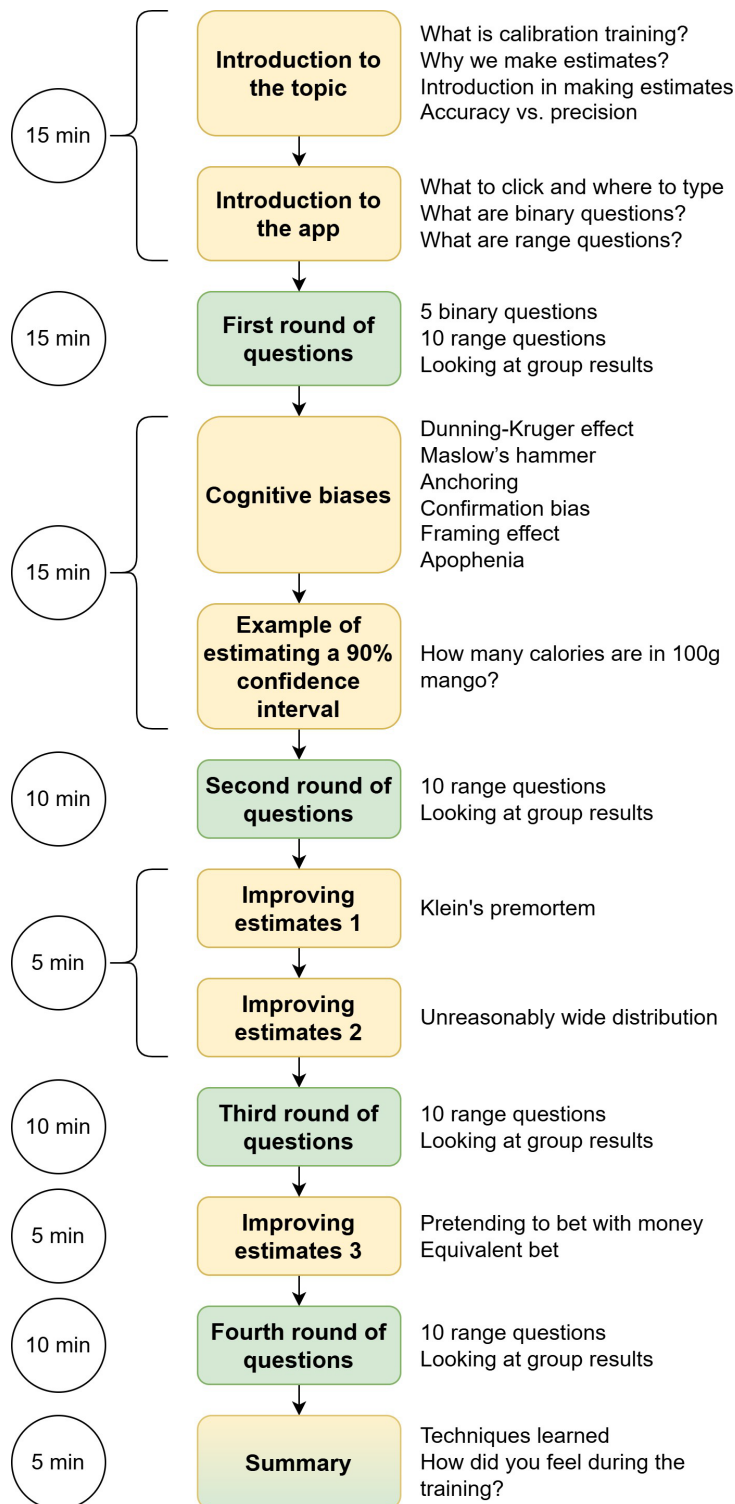


Figure 4: Flow chart of a calibration training session with four rounds of questions. The circles on the left indicate the duration of each step (90 minutes total), with yellow boxes for instructor-led input and green boxes for participant activities. The right side lists the content of each step.

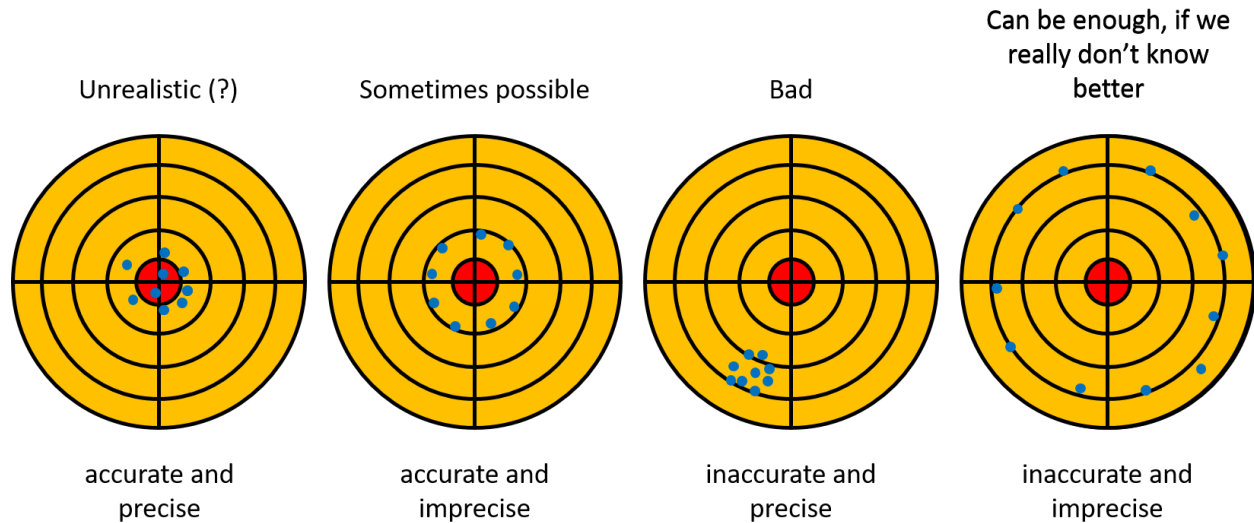


Figure 5: Precision versus accuracy explained with the example of shooting arrows at a target.

Understanding Cognitive Biases:

We introduce several cognitive biases to help participants recognize factors that may distort judgment:

- *Dunning-Kruger effect*: The overconfidence of individuals with little experience (Kruger & Dunning, 1999).
- *Maslow's Hammer*: A preference for familiar tools (Maslow, 1966).
- *Anchoring effect*: The influence of unrelated numbers on estimates, illustrated through Kahneman's rigged wheel experiment (2011). Additional biases, such as confirmation and framing effects, are also discussed to deepen understanding of these psychological influences on estimation.

Practical Estimation Exercise:

To introduce estimation techniques, we ask participants to estimate, for example, the caloric content of 100g of mango. This exercise illustrates various estimation types and demonstrates the construction of 90% confidence intervals—ranges expected to contain the correct answer 90% of the time.

Table 1: Estimation types are explained with the example of the question, "How many calories are in 100g of mango?"

Estimation type	Estimated value	Comment
Bad estimate	200 kcal	no range, almost certainly wrong
Non-informative estimate	0-1000 kcal	definitely right, but not very useful
Good non-expert estimate	20-300 kcal	accurate, but not very precise
Good expert estimate	40-150 kcal	accurate and quite precise

Techniques to improve estimation skills:

To enhance estimation accuracy, we incorporate several techniques (Table 2).

Table 2: Techniques to improve estimation skills.

Name	Approach	Reference
Klein's premortem	Prospective hindsight—imagining that an event has already occurred and led to an undesirable outcome. Ask for the value of lower and upper bound, and explore why this bound could be wrong i.e., why could the truth be outside the range	(Klein, 2007)
Unreasonably wide distribution	Begin with a broad range and narrow it to reflect a 90% confidence interval	Own concept by Whitney and Luedeling
Pretending to bet with money	Just imagine that you could win some money if your range includes the true value	(Hubbard, 2014)
Equivalent bet	Imagine you could win 1000\$. Would you prefer spinning a dial with 90% chance to win or to bet on your estimated interval? If your interval reflects your 90% confidence interval you should be indifferent between the options. If you prefer the dial you should widen your range, but if you prefer your estimated range, you should narrow it.	(Hubbard, 2014)

Question Types and Training Flow

In a typical training session, we use both binary and range questions. Binary questions (true/false) serve as an entry point, helping participants build confidence in estimating and establishing a grounding in calibration. Range questions, which require estimating intervals, engage participants in a more complex reflection on uncertainty, reinforcing the real-world applications of accurate estimation.

Binary questions (Figure 6) allow participants to gauge their confidence on a sliding scale from 50% (complete uncertainty) to 100% (absolute certainty), helping them recognize initial biases.

Question:1

One year contains 687 days on Mars.

Response:

☒ TRUE

☐ FALSE

Confidence:

50% 60% 100%

50 55 60 65 70 75 80 85 90 95 100

→ Next

Figure 6: Example binary question with confidence slider, from the calibration app.

Range questions (Figure 7) ask participants to provide a 90% confidence interval for numeric answers. These questions require deeper engagement, using introduced techniques to narrow down ranges effectively over rounds.

Question:2

How many liters of water does the average household in the United States use per day?

90% Confidence Interval

Lower Bound **Upper Bound**

0 0

→ Next

Figure 7: Example range question with confidence slider, from the calibration app.

We have subjected participants across several international projects to calibration training with the app.

RESULTS

In our recent calibration training sessions using the app, we have gathered insightful preliminary results and valuable user feedback. Participants have reported that the app effectively supports them in understanding their own biases and improving the accuracy of their confidence

intervals. For example, users noted a heightened awareness of their tendencies to either overestimate or underestimate uncertainty, and they appreciated the app's interactive nature in addressing these cognitive biases.

The pilot testing suggests several anticipated benefits of this training approach. First, we observe an overall improvement in the accuracy of confidence intervals provided by participants, indicating enhanced calibration of uncertainty estimates. Second, by using the app, participants gained better self-awareness of their individual biases, which helped to reduce systematic errors in their judgments. The practical application of these improvements is expected to enhance decision-making quality, particularly in complex environments where uncertainty plays a crucial role.

These preliminary insights are promising, and we are optimistic that continued use of the app, combined with workshops and iterative practice, will further support participants in mastering calibration techniques, ultimately leading to more informed and confident decision-making.

DISCUSSION

Preliminary Results and Feedback

Our pilot testing has involved diverse groups across several projects, providing valuable insights into the effectiveness of the calibration training tool. These projects span different sectors and geographic regions, allowing us to test the app's adaptability and impact in real-world contexts.

In the NIFAM project, which focuses on nutrition interventions in Vietnam and Myanmar, participants highlighted the app's utility in enhancing their ability to make reliable uncertainty estimations. Models have been developed and fed with calibrated estimates to inform decisions about effective nutrition programs in complex settings (Whitney et al. 2024).

The "Experimentierfeld Südwest" digitalization initiative in Rhineland-Palatinate focuses on decisions on late frost protection and 2-D fruiting wall training systems, as well as an apple yield and quality forecast tool. The app has proved helpful in generating models for decision-making in fruit orchard management (Schmitz et al. 2023).

The INTERFACES project is applying calibration training for models supporting sustainable land management pathways in Africa. The app has been applied to help stakeholders—ranging from farmers to policymakers—develop a deeper understanding of uncertainty, facilitating more informed decision-making about sustainable practices (Sanginga et al. 2024).

The REFOREST and ABCDR projects both focus on overcoming barriers to agroforestry adoption in the European Union. In the REFOREST project, the app supported participants in estimating complex variables like biodiversity improvements, soil enhancement, and carbon sequestration, which are crucial for agroforestry decision-making. Similarly, in the ABCDR project, part of the CDRterra funding measure aimed at supporting Germany's greenhouse gas emission neutrality by 2045, participants found the calibration tool instrumental in evaluating the effectiveness of carbon dioxide removal technologies in agroforestry systems. Both projects highlighted the app's potential in enhancing decision quality under uncertainty for land-use management and climate mitigation strategies.

The HortiPrimed project, focused on supporting non-chemical plant protection in horticultural production, also used the app for calibration training. Participants found that using the app helped them refine their confidence intervals when estimating factors related to yield and quality under changing climatic conditions, ultimately improving non-chemical intervention strategies (Moon et al. 2024).

Overall, participant feedback across these projects has been positive, particularly regarding the structured learning environment and the app's focus on recognizing and mitigating cognitive biases. Users reported feeling more confident in providing accurate estimates, attributed to the app's targeted exercises and real-time feedback.

Anticipated Benefits

Based on the pilot testing and theoretical foundations, the use of the calibration training tool yields several key benefits. First, participants develop improved estimation accuracy, especially by constructing 90% confidence intervals. The app's interactive exercises help internalize these skills, reducing systematic errors such as overconfidence and anchoring. Additionally, integrating cognitive bias awareness training positively impacts decision quality in the long term, particularly in high-stakes contexts like agricultural planning, land management, and climate-sensitive production systems.

The collaborative nature of the training also encourages shared learning among participants, enriching their understanding of uncertainty and its practical implications. This is particularly beneficial in diverse and interdisciplinary teams, such as those in our pilot projects, where combining expertise from different domains is essential for effective decision-making.

Weaknesses of the approach

While our calibration training tool has shown promising results, certain limitations must be addressed, especially when working in diverse contexts, such as with illiterate communities or in settings where participants may not have adequate time to engage with the tool extensively.

One major challenge is working with participants with limited literacy or technical skills, which makes interacting with the web-based interface difficult. In these cases, we have employed interpreters to interface the tool and the participants. This approach has proven effective, particularly in Uganda, as demonstrated in Whitney et al. (2018) where interpreters facilitated the calibration training for local experts. However, for this method to succeed, the interpreter must be well-calibrated and comfortable with the tools being used. They must be capable not only of interpreting terms and ideas but also of offering interpretation of the estimates provided, as well as managing the interface. This level of involvement requires considerable training, which is not always feasible. Additionally, relying on interpreters can add complexity and potential errors to the estimation process.

Another notable weakness is the issue of expert fatigue and boredom during the calibration process. Once experts have become calibrated and are proficient in estimating uncertainties, they may lose interest when faced with a long list of variables requiring estimates. This can reduce the accuracy and quality of the estimates, as experts may rush through the process or disengage. To address this, we propose streamlining the interface to allow for quick estimate gathering. We also propose leveraging other methods for generating estimates where feasible. Many models and methods exist that provide reasonable values for various variables. Instead of

repeatedly calibrating experts for each variable, we could gather uncertainty distributions from existing model results from past calibration processes and also make use of these. This approach could significantly streamline the process, reduce the burden on experts, and maintain the quality of the estimates. However, we should carefully consider how these data are generated and how appropriate they are for differing geographic and socio-economic conditions.

CONCLUSIONS

Our online calibration app is a structured, versatile platform designed to enhance estimation skills by integrating principles from cognitive psychology and decision analysis. The app's dual-interface supports dynamic interactions between participants and analysts, enabling immediate, relevant feedback. Users gain awareness of cognitive biases through repeated engagement with binary and range questions, allowing for more informed, confident decision-making.

This approach holds broad applicability across fields, including academia, policy, and industry, where accurate estimation underpins successful outcomes. Empowering individuals with robust calibration skills contributes to an effective decision-making framework that addresses complex challenges. Future directions for this work include incorporating other types of questions beyond binary and range formats, exploring settings beyond traditional calibration training, and developing more adaptive interfaces to better serve the needs of diverse user groups. Expanding the scope of the app to include different types of decision-making scenarios, such as resource allocation or risk assessment, could further enhance its utility and impact.

We encourage ongoing development of this training methodology within the scientific community for broader applications and improvements.

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